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DOMED THERMOSTAT LENS ASSEMBLY

Abstract

Various arrangements of thermostats are detailed herein. A thermostat can include a housing, an electronic display, an ambient light sensor, and a lens assembly. The lens assembly can include a circular lens. The circular lens can include a first surface that is flat. A perimeter of the circular lens can be defined by a second surface that is perpendicular to the first surface of the circular lens. Third surface of the circular lens may be domed. The ambient light sensor can measure ambient light within an ambient environment of the thermostat through the domed circular lens.

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Background/Summary

BACKGROUND

[0001] Thermostats tend to be highly visible products in persons' homes. Further, where heating and cooling wiring are installed can dictate where a thermostat is installed within a home. Thermostats are therefore required to be functional as well as aesthetically pleasing. Embodiments detailed herein can accomplish both of these goals.

SUMMARY

[0002] Various embodiments are described related to a thermostat. In some embodiments, a thermostat is described. The thermostat may comprise a housing. The thermostat may comprise an electronic display housed by the housing. The thermostat may comprise an ambient light sensor housed by the housing. The thermostat may comprise a lens assembly attached with the housing. The lens assembly may comprise a circular lens. A first surface of the circular lens may be flat. A perimeter of the circular lens may be defined by a second surface that may be perpendicular to the first surface of the circular lens. A third surface of the circular lens may be domed. The ambient light sensor may measure ambient light within an ambient environment of the thermostat through the circular lens.

[0003] Embodiments of such a device may include one or more of the following features: a size of a field of view of the ambient light sensor may be increased by the third surface of the circular lens. The lens assembly may further comprise a polarized mirrored film attached with the circular lens. The polarized mirrored film may be oriented with respect to the electronic display to increase transmissivity of polarized light emitted by the electronic display. The lens assembly may further comprise a colored ink layer attached between the polarized mirrored film and the circular lens. The lens assembly may further comprise a masking layer attached between the polarized mirrored film and the electronic display. The lens assembly may further comprise one or more adhesive layers and only the one or more adhesive layers secure the lens assembly to the thermostat. The device may further comprise a radar sensor that emits radar signals and detects reflections of the radar signals through the lens assembly. The circular lens may comprise polymethyl methacrylate (PMMA). The circular lens may be formed using injection compression molding. The second surface of the circular lens may be completely surrounded by a rotatable ring.

[0004] In some embodiments, a method for manufacturing a thermostat lens assembly is described. The method may comprise creating a domed lens. A first surface of the circular lens may be flat. A perimeter of the circular lens may be defined by a second surface that may be perpendicular to the first surface of the circular lens. A third surface of the circular lens may be domed. The method may comprise measuring a direction of polarization of a mirror film. The method may comprise determining an orientation of the mirror film to attach with a lens assembly comprising the domed lens based on the measured direction of polarization of the mirror film. The method may comprise installing the lens assembly in a thermostat such that the determined orientation minimizes a polarizing effect on light emitted by an electronic display of the thermostat.

[0005] Embodiments of such a method may include one or more of the following features: the method may further comprise affixing a masking layer with the mirror film based on the determined orientation. The masking layer may be asymmetrical. The domed lens may be created using injection compression molding. The domed lens may be created from polymethyl methacrylate (PMMA). The method may further comprise affixing the circular lens to a colored ink layer using an optically clear adhesive. The method may further comprise affixing the mirrored film with the colored ink layer. The method may further comprise affixing one or more adhesive layers to the masking layer. Only the one or more adhesive layers may secure the lens assembly to the thermostat.

[0006] In some embodiments, a thermostat lens assembly comprising a circular lens is described. A first surface of the circular lens may be flat. A perimeter of the circular lens may be defined by a second surface that may be perpendicular to the first surface of the circular lens. A third surface of

the circular lens may be domed. The device may further comprise a mirrored film layer having a polarization. The device may further comprise a masking layer that may be asymmetrical. The mirrored film may be oriented with respect to the masking layer based on the polarization. The device may further comprise an optically clear adhesive layer. The device may further comprise a colored ink layer. The optically clear adhesive layer may adhere the lens to the colored ink layer and the mirrored film layer may be in direct contact with the colored ink layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A further understanding of the nature and advantages of various embodiments may be realized by reference to the following figures. In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0008] FIG. 1 is a block diagram of an embodiment of a smart thermostat system.

[0009] FIG. 2A is an isometric view of an embodiment of a smart thermostat.

[0010] FIG. 2B is a front view of an embodiment of a smart thermostat.

[0011] FIG. 2C is a side view of an embodiment of a smart thermostat.

[0012] FIG. 3A is an exploded front isometric view of an embodiment of a thermostat.

[0013] FIG. 3B is an exploded rear isometric view of an embodiment of a thermostat.

[0014] FIG. 4 is an exploded front isometric view of an embodiment of the layers of a domed lens assembly.

[0015] FIG. 5 is a cross section of an embodiment of a thermostat.

[0016] FIG. 6A illustrates a cross section of an embodiment of a lens assembly.

[0017] FIG. 6B illustrates a cross section of an embodiment of a lens assembly which alters the field of view of an ambient light sensor.

[0018] FIG. 7 illustrates an embodiment for manufacturing a domed lens assembly for a thermostat.

DETAILED DESCRIPTION

[0019] A thermostat may have a domed lens that functions as part of a lens assembly. Since an electronic display may emit light through the domed lens and since the lens assembly may have a mirrored effect, any defects in the domed lens would be readily visible. In order to make the domed lens as defect-free as possible, injection compression molding can be used to manufacture the domed lens. In addition to being aesthetically pleasing, such a domed lens can help increase a field of view of one or more sensors, such as an ambient light sensor, that senses the environment through the lens assembly.

[0020] The lens assembly can include multiple layers to achieve desired visual effects. A domed lens can be attached with multiple layers including a colored ink layer, a mirror film layer, and/or a masking layer. The mirror film layer may be polarized. The orientation of polarization may need to be accounted for during manufacture to prevent excessive blocking of light emitted by an electronic display. To do so, the polarization of the mirror film layer may be determined such that the mirror film layer can be oriented with respect to the electronic display of the thermostat to maximize transmissivity of the mirror film layer for light emitted by the display.

[0021] The figures and following description provide additional detail. FIG. 1 is a block diagram of an embodiment of a smart thermostat system. Smart thermostat system **100** can include smart thermostat **110**; backplate **120**; heating, ventilation, and air conditioning system (HVAC) **125**; wall

plate **130**; network **140**; cloud-based server system **150**; and computerized device **160**. Smart thermostat **110** represents embodiments of the thermostats detailed herein. Thermostat **110** can include: electronic display **111**; user interface **112**; radar sensor **113**; network interface **114**; speaker **115**; ambient light sensor **116**; one or more temperature sensors **117**; HVAC interface **118**; processing system **119**; housing **121**; and lens assembly **122**.

[0022] Electronic display **111** may be visible through lens assembly **122**. In some embodiments, electronic display **111** is only visible when electronic display **111** is at least partially illuminated. In some embodiments, electronic display **111** is a touch screen which can allow electronic display **111** to serve as a user interface to receive input. If a touch screen, electronic display **111** can allow one or more gestures, including tap and swipe gestures, to be detected.

[0023] User interface **112** can be various forms of input devices through which a user can provide input to smart thermostat **110**. In some embodiments herein, an outer rotatable ring is present as part of user interface **112**. The ring can be rotated by a user clockwise and counterclockwise in order to provide input. The ring can be infinitely rotatable in either direction, thus allowing a user to scroll or otherwise navigate user interface menus. The ring (and, possibly, lens assembly **122**) can be pressed inward (toward the rear of smart thermostat **110**) to function as a “click” or to make a selection. The outer rotatable ring can, for example, allow the user to make temperature target adjustments. By rotating the outer ring clockwise, the target temperature can be increased, and by rotating the outer ring counterclockwise, the target temperature can be decreased. As another example, the ring can be rotated to highlight displayed icons; an inward click can be provided by a user to select a particular icon.

[0024] Radar sensor **113** may be configured to output electromagnetic radiation into the ambient environment in front of electronic display **111** of thermostat **110**. Radar sensor **113** may be an integrated circuit that includes one or more antennas, one or more RF emitters, and one or more RF receivers. Radar sensor **113** may be able to detect the presence of a user and the distance at which the user is located. Other details of a user may also be detected, such as velocity and whether the user is facing smart thermostat **110**. Radar sensor **113** may use frequency-modulated continuous wave (FMCW) radar. Radar sensor **113** may emit electromagnetic radiation and receive reflected electromagnetic radiation through lens assembly **122**. Radar sensor **113** may emit chirps of radar that sweep from a first frequency to a second frequency. Therefore, the waveform output by radar sensor **113** may be a sawtooth waveform. Using receive-side beam steering on the reflected radio waves received using multiple antennas, certain regions may be targeted for sensing the presence of users. For instance, beam steering away from the ground may be performed to avoid pets being potentially incorrectly detected as a user.

[0025] Network interface **114** may be used to communicate with one or more wired or wireless networks. Network interface **114** may communicate with a wireless local area network, such as a Wi-Fi network. Additional or alternative network interfaces may also be present. For example, smart thermostat **110** may be able to communicate with a user device directly, such as using Bluetooth or some other device-to-device short-range wireless communication protocol. Thermostat **110** may be able to communicate via a mesh network with various other home automation devices, such as using Thread. Mesh networks may use relatively less power compared to wireless local area network-based communication, such as Wi-Fi. In some embodiments, thermostat **110** can serve as an edge router that translates communications between a mesh network and a wireless local area network, such as a Wi-Fi network. In some embodiments, a wired network interface may be present, such as to allow communication with a local area network (LAN). One or more direct wireless communication interfaces may also be present, such as to enable direct communication with a remote temperature sensor installed in a different housing external and distinct from housing **121**. The evolution of wireless communication to fifth generation (5G) and sixth generation (6G) standards and technologies provides greater throughput with lower latency which enhances mobile broadband services. 5G and 6G technologies also provide new classes of

services, over control and data channels, for vehicular networking (V2X), fixed wireless broadband, and the Internet of Things (IoT). Thermostat **110** may include one or more wireless interfaces that can communicate using 5G and/or 6G networks.

[0026] Speaker **115** can be used to output audio. Speaker **115** may be used to output beeps, clicks, synthesized speech, or other audible sounds, such as in response to the detection of user input via user interface **112**.

[0027] Ambient light sensor **116** may sense the amount of light present in the environment of thermostat **110**. Measurements made by ambient light sensor **116** may be used to adjust the brightness of electronic display **111**. In some embodiments, ambient light sensor **116** senses an amount of ambient light through lens assembly **122**. Therefore, compensation for the reflectivity of lens assembly **122** may be made such that the ambient light levels are correctly determined via ambient light sensor **116**. In some embodiments, a light pipe is present between ambient light sensor **116** and lens assembly **122** such that, in a particular region of lens assembly **122**, light that is transmitted through lens assembly **122** is directed to ambient light sensor **116**, which may be mounted to a printed circuit board (PCB), such as a PCB to which processing system **119** is attached.

[0028] One or more temperature sensors **117** may be present within thermostat **110**. Temperature sensors **117** may be used to measure the ambient temperature in the environment of thermostat **110**. One or more additional temperature sensors that are remote from thermostat **110** may additionally or alternatively be used to measure the temperature of the ambient environment.

[0029] Lens assembly **122** may have a transmissivity sufficient to allow illuminated portions of electronic display **111** to be viewed through lens assembly **122** from an exterior of thermostat **110** by a user. Lens assembly **122** may have a reflectivity sufficient such that portions of lens assembly **122** that are not illuminated from behind appear to have a mirrored effect to a user viewing a front of thermostat **110**. Further detail regarding lens assembly **122** are provided in relation to FIGS. 4-7.

[0030] HVAC interface **118** can include one or more interfaces that control whether a circuit involving various HVAC control wires that are connected either directly with thermostat **110** or with backplate **120** is completed. A heating system (e.g., furnace, boiler, heat pump), cooling system (e.g., air conditioner, heat pump), and/or fan may be controlled via HVAC wires by opening and closing circuits that include the HVAC control wires. In some installations, only a heating system or cooling system is controlled by smart thermostat **110**; in other embodiments, smart thermostat **110** may control both a heating system and cooling system.

[0031] Processing system **119** can include one or more processors. Processing system **119** may include one or more special-purpose or general-purpose processors. Such special-purpose processors may include processors that are specifically designed to perform the functions detailed herein. Such special-purpose processors may be ASICs or FPGAs which are general-purpose components that are physically and electrically configured to perform the functions detailed herein. Such general-purpose processors may execute special-purpose software that is stored using one or more non-transitory processor-readable mediums, such as random access memory (RAM), flash memory, a hard disk drive (HDD), or a solid state drive (SSD) of thermostat **110**.

[0032] Processing system **119** may output information for presentation to electronic display **111**. Processing system **119** can receive information from temperature sensors **117**, user interface **112**, radar sensor **113**, network interface **114**, and ambient light sensor **116**. Processing system **119** can perform bidirectional communication with network interface **114**. Processing system **119** can output information to be output as sound to speaker **115**. Processing system **119** can control the HVAC system via HVAC interface **118**.

[0033] Housing **121** may house and/or attach with all of the components of smart thermostat **110**, either directly or via other components. For example, lens assembly **122** may adhere to electronic display **111**, which is attached with housing **121**.

[0034] Thermostat **110** may be attached (and removed) from backplate **120**. Some number of

HVAC control wires may be attached with terminals or receptacles of backplate **120**. Such HVAC control wires electrically connect backplate **120** with HVAC system **125**, which can include a heating system, cooling system, ventilation system, or some combination thereof. Backplate **120** can allow smart thermostat **110** to be attached and removed from backplate **120** without affecting the electrical connections of the HVAC control wires with backplate **120**. In other embodiments, such control wires are directly connected with smart thermostat **110**. In some embodiments, wall plate **130** may additionally be installed between backplate **120** and a surface, such as a wall, such as for aesthetic reasons (e.g., cover an unsightly hole through which HVAC wires protrude from the wall.)

[0035] Network **140** can include one or more wireless networks, wired networks, public networks, private networks, and/or mesh networks. A home wireless local area network (e.g., a Wi-Fi network) may be part of network **140**. Network **140** can include the Internet. Network **140** can include a mesh network, which may include one or more other smart home devices, may be used to enable thermostat **110** to communicate with another network, such as a Wi-Fi network. Thermostat **110** may function as an edge router that translates communications from a relatively low power mesh network received from other devices to another form of network, such as a relatively higher power network, such as a Wi-Fi network.

[0036] Cloud-based server system **150** can maintain an account mapped to smart thermostat **110**. Thermostat **110** may periodically or intermittently communicate with cloud-based server system **150** to determine whether setpoint or schedule changes have been made. A user may interact with thermostat **110** via computerized device **160**, which may be a mobile device, smartphone, tablet computer, laptop computer, desktop computer, or some other form of computerized device that can communicate with cloud-based server system **150** via network **140** or can communicate directly with thermostat **110** (e.g., via Bluetooth or some other device-to-device communication protocol). A user can interact with an application executed on computerized device **160** to control or interact with thermostat **110**.

[0037] FIG. 2A is an isometric view of an embodiment of a smart thermostat **200**. Smart thermostat **200** can represent an embodiment of smart thermostat **110** of FIG. 1. In FIG. 2A, electronic display **202**, located behind lens assembly **212**, is active in displaying a setpoint temperature. The housing of smart thermostat **200** can define sidewall **208**. Sidewall **208** may be generally cylindrical according to various embodiments. A diameter of the sidewall **208** may be smaller than a diameter of the electronic display **202** and ring **210** according to various embodiments and as illustrated in FIG. 2A. Ring **210** can function as detailed in relation to user interface **112**. Either attached with housing **121** or attached with components connected with housing **121** is lens assembly **212**. Lens assembly **212** may include a reflective layer having a reflectivity such that when the electronic display **202** is not illuminated, lens assembly **212** appears to be a mirror when viewed by a user.

[0038] In some embodiments, ring **210** is mounted to lens assembly **212**. In other embodiments, ring **210** can be rotated clockwise and counterclockwise independent of lens assembly **212**. In some embodiments, housing **121** includes a display frame (not visible in this view) that further supports electronic display **202** and lens assembly **212**.

[0039] Electronic display **202** is housed behind lens assembly **212** such that, when illuminated, the portion of electronic display **202** that is illuminated is visible through lens assembly **212** by a user. In some embodiments, due to the reflectivity of lens assembly **212**, an edge of electronic display **202** is not visible to a user regardless of whether electronic display **202** is illuminated, partially illuminated, or not illuminated. Therefore, the overall effect experienced by a user may be that lens assembly **212** appears as a mirror and portions of electronic display **202**, when illuminated, are visible through lens assembly **212**.

[0040] In various embodiments, around an axis perpendicular to the display face of electronic display **202**, the ring **210** has an inner diameter and an outer diameter and both the inner diameter and the outer diameter of ring **210** are larger than a diameter of sidewall **208** of housing **121**.

[0041] FIG. 2B is a front view of an embodiment of smart thermostat **200**. When mounted on a wall or other surface, lens assembly **212** is opposite the portion of smart thermostat **200** that mounts to the wall or other surface. Therefore, when a user is facing mounted smart thermostat **200**, lens assembly **212** is visible. Lens assembly **212** can form an uninterrupted circular surface with no gaps, holes, lens, or other discontinuities present on the outermost surface of lens assembly **212**. Lens assembly **212** has sufficient transmissivity to allow light emitted by electronic display **202** located within housing **206** to be visible through lens assembly **212**. Further, lens assembly **212** may have sufficient reflectivity such that a mirrored effect is present on portions of lens assembly **212** that are not currently being illuminated from behind by electronic display **202**.

[0042] Present in FIG. 2B is an indication of cross-section **500**. Cross-section **500** is detailed in relation to FIG. 5.

[0043] FIG. 2C is a side view of an embodiment of a smart thermostat. When smart thermostat **200** is mounted to a wall or other surface, sidewall **208** of housing **121** is visible. Around an axis **250**, the ring **210** has an inner diameter D_i and an outer diameter D_o and both the inner diameter D_i and the outer diameter D_o of the ring **210** are larger than a diameter D_h of sidewall **208** of housing **121**. According to various embodiments, sidewall **208** of housing **121** can be generally cylindrical and can have a consistent diameter along a length thereof. Alternatively, a diameter of sidewall **208** can increase as a distance from lens assembly **212** increase.

[0044] In some embodiments, ring **210** has a smallest diameter at the rearmost portion of ring **210**. D_r is indicative of the diameter of ring **210** where ring **210** meets sidewall **208**. This arrangement can help facilitate a user's fingers reaching around ring **210**, grasping ring **210**, and rotating in either direction. In some embodiments, along axis **250**, sidewall **208** may have a diameter of approximately D_r wherein ring **210** and sidewall **208** meet. In some embodiments, the diameter of sidewall **208** can increase as the distance from ring **210** increases.

[0045] FIG. 3A is an exploded front isometric view of an embodiment of smart thermostat **200**. FIG. 3B is an exploded rear isometric view of smart thermostat **200**. Viewing the components of the smart thermostat **200** left to right, lens assembly **212** forms an outermost domed surface of smart thermostat **200**. Adjacent lens assembly **212** may be electronic display **202**. Electronic display **202** may be a liquid-crystal display (LCD) or organic light emitting diode (OLED) display according to various embodiments. In at least some embodiments, one or more adhesives may be used to attach electronic display **202** with lens assembly **212**. An exploded view of lens assembly **212** is provided in relation to FIG. 4.

[0046] According to at least some embodiments, electronic display **202** is supported by a display frame **302**. Smart thermostat **200** further includes one or more antenna assemblies **304** for communicating with a network and/or other electronic devices. Antenna assembly **304** can be used for communicating with wireless local area networks (e.g., Wi-Fi), device-to-device communication (e.g., Bluetooth), and/or communicating with mesh networks (e.g., Thread). Smart thermostat **200** includes one or more sensor boards, such as sensor daughterboard **306**. One or more temperature sensors may be installed on sensor daughterboard **306**. Use of sensor daughterboard **306** can help isolate the one or more temperature sensors from heat generated by other components.

[0047] Smart thermostat **200** may further include clip **308** for coupling ring **210** and display frame **302** supporting electronic display **202**. Clip **308** may act as an axial constraint for smart thermostat **200**. In particular, clip **308** prevents electronic display **202**, display frame **302**, and ring **210** from decoupling from one another in the assembled configuration.

[0048] As shown in FIG. 3A, smart thermostat can include magnetic strip **310**. According to various embodiments, ring **210** rotates relative to sidewall **208** of housing **206** and a backplate when smart thermostat **200** is mounted to a surface. In various embodiments, a sensor installed on a sensor board, such as sensor board **306** and magnetic strip **310** are used for detecting rotation of the ring **210** during use.

[0049] According to various embodiments, ring **210** is mounted to housing **206** such that ring **210**

can be rotated clockwise and counterclockwise. Ring **210** may include polished stainless steel and a finish applied using physical vapor deposition (PVD). Ring **210** further advantageously provides an aesthetic appearance as the finish of the ring **210** appears seamless relative to lens assembly **212** having a mirrored effect.

[0050] Further internal components of smart thermostat **200** include battery **312** and battery adhesive **314**. Battery **312** can be a secondary battery and can provide power to the various components of smart thermostat **200**, including electronic display **202** and processing system **119**. Battery adhesive **314** may be used to adhere battery **312** within housing **206** although the battery **312** (or any other components of the smart thermostat **200**) may be secured within the housing **206** using other means. For example, various components may be secured using adhesives, screws, wires, clips, or the like.

[0051] Smart thermostat **200** includes processing system **316**. According to some embodiments, processing system **316** is a system-on-a-chip (SoC) including various processing parts, memory, modems, etc. Processing system **316** may be in electric communication with one or more antennas present on antenna assembly **304**, sensor board **306**, electronic display **202**, etc., for performing various functions of the smart thermostat **200** and outputting results based on user input (e.g., in response to the user rotating the ring **210** and/or user input via an external mobile device). Adjacent processing system **316** may be piezo sensor **317**. Additional components of the processing system **316** or components that work with processing system **316** are also shown in FIG. 3. For example, multi-layer board (MLB) **318** may be provided for performing various functions of smart thermostat **200**, in a manner that would be appreciated by one having ordinary skill in the art. In some embodiments, MLB **318** may include a Universal Serial Bus (USB) port for electrically coupling smart thermostat **200** to another electronic device for various updates, servicing, or the like. Various springs **319** for supporting components, flexes **321** for enabling flexible and high-density interconnects between printed circuit boards (PCBs), LCDs, etc., and additional links **323** may also be included in the internal components of smart thermostat **200**.

[0052] Smart thermostat **200** may include more or fewer components than those shown in FIG. 3A and FIG. 3B. In various embodiments, the components may be in one or more configurations other than the configuration shown in FIG. 3A. Advantageously, various components of smart thermostat **200** are optimized to be condensed into housing **206** such that the overall side profile of smart thermostat **200** is significantly thinner than a side profile of other commercially available smart thermostats.

[0053] FIG. 4 is an exploded front view of various embodiments of lens assembly **400**. Lens assembly **400** can represent embodiments of lens assembly **122** and **212**. In particular, FIG. 4 illustrates an embodiment of a stack of components that can be used to create lens assembly **122**. Lens assembly **400** can include: domed lens **402**; optically clear adhesive (OCA) layer **404**; tinted ink layer **406**; mirror film **408**; masking layer **410**; frame pressure sensitive adhesive (PSA) **412**; and display PSA **414**. While embodiments of lens assembly **400** may be used on smart thermostat **200**, embodiments of such a lens assembly may be used on other forms of smart devices. For instance, lens assembly **400** can be incorporated as part of a smart assistant device or a smart watch.

[0054] Domed lens **402** may be domed on an outer surface and flat on an inner surface that is in contact with OCA layer **404**. Further detail regarding the shape of domed lens **402** is provided in reference to FIG. 5. Domed lens **402** can be formed from polymethyl methacrylate (PMMA), which can provide a transparency similar to glass. Other plastic or acrylic materials are also possible. Domed lens **402** may also be formed from glass. Domed lens **402** can be formed using injection compression molding. Injection compression molding can be used because it allows for defect-free surfaces to be formed. To perform injection compression molding of domed lens **402**, material can be injected into a nearly closed mold. The mold may then be compressed such that the injected material conforms to the shape of the mold. Excess material can be removed, such as

through machining.

[0055] Domed lens **402** is circular and does not have any holes, vents, gaps, or other discontinuities present on it. Similarly, no holes, vents, gaps, or other discontinuities are present on at least OCA layer **404**, tinted ink layer **406**, and mirror film layer **408**. Having continuous material helps to maintain a consistent visual effect across the entirety of lens assembly **400** as viewed by a user.

[0056] OCA layer **404** can be a pressure or temperature sensitive adhesive that adheres domed lens **402** with tinted ink layer **406**. Tinted ink layer **406** can be a transparent layer that tints light passing through tinted ink layer **406**. Since tinted ink layer **406** is closer to domed lens **402** than mirror film layer **408**, both light by mirror film layer **408** and light emitted by electronic display **111** is tinted. The color used for tinting can be selected based on aesthetics.

[0057] Mirror film layer **408** may have sufficient reflectivity that when electronic display **111** is not illuminated, a user viewing lens assembly **400** may see a reflection of himself, herself, or the ambient environment. For example, mirror film layer **408** can be Toray® 125FH-40 mirror film. Mirror film layer **408** may be polarized. Due to the way some mirror films are manufactured, throughout a roll of mirror film, the direction of polarization can vary. When a piece of mirror film is stamped or cut out to form mirror film layer **408**, the direction of polarization may be determined in order to orient in relation the electronic display, which also outputs polarized light. If orientation is not controlled, visibility of the electronic display through mirror film layer **408** may be adversely affected. Further detail regarding orientation of mirror film layer **408** is detailed in relation to FIG. 7.

[0058] Masking layer **410** can be used to block a user from viewing components blocked by the opaque portions of masking layer **410**. Masking layer **410** may be black or another dark color to make it difficult to see through mirror film layer **408**. Masking layer **410** can obscure a view of frame adhesive **414** and display adhesive **416**. Masking layer **410** may be asymmetric. Therefore, it must be oriented in a particular orientation with respect to other components of smart thermostat **200**. For example, masking layer **410** includes a hole for an ambient light sensor to have a field of view of the ambient environment through domed lens layer **402**, OCA layer **404**, tinted link layer **406**, and mirror film layer **408**.

[0059] Furthermore, the masking layer **410** may help enhance the effect that the electronic display is seamless with lens assembly **400**. A color value for masking layer **410** may be selected, having an appropriate lightness value, such that it is difficult or impossible for a user to visually see an edge of the electronic display screen within the smart device. By obscuring an edge of the edge of the electronic display, a user may have the impression that the entire region behind domed lens **402** is electronic display **111**.

[0060] Obscured behind masking layer **410** may be two separate adhesive layers. Frame adhesive layer **412** may adhere domed lens layer **402**, OCA layer **404**, tinted link layer **406**, mirror film layer **408**, and masking layer **410** to display frame **302**. Display adhesive layer **414** may adhere domed lens layer **402**, OCA layer **404**, tinted link layer **406**, mirror film layer **408**, and masking layer **410** to electronic display **202**. Different types of adhesives may be used to provide better adhesion to the material of electronic display **202** and display frame **302**. Adhesive layer **412** and display adhesive layer **414** may both be different types of pressure sensitive adhesives (PSAs). In other embodiments, a single adhesive layer may be used. For example, 3M® 5126-025 may be used as the PSA.

[0061] FIG. 5 is a cross section **500** of an embodiment of smart thermostat **200**. The location and direction of cross section **500** is indicated on FIG. 2B. The domed profile of domed lens **402** is visible in the cross section of FIG. 5. Surface **501** is the outer surface of domed lens **402** that is adjacent the ambient environment and which a user can touch. An entirety of surface **501** is convex from edge to edge. Surface **502** is the inner surface and adheres with OCA layer **404**. OCA layer **404** and other layers of lens assembly **400** are not visible in FIG. 5. An entirety of surface **502** can be flat. Surface **503** forms a circumference around the entirety of domed lens **402**. Surface **503** is

perpendicular or approximately perpendicular (defined as within 5° of perpendicular) to surface **502**.

[0062] Electronic display **202** is disposed under the domed lens **402** and surrounded by rotatable ring **510**. In particular, ring **210** surrounds surface **503** of domed lens **402** and couples to housing **206**, which has a cylindrical sidewall **208**.

[0063] FIG. **6A** illustrates a cross section **600A** of an embodiment of lens assembly **212**. Cross section **600A** of FIG. **6A** is not to scale. Cross section **600A** represents a cross section at the same location as cross section **500**, but only including lens assembly **212**. Domed lens **402** has center thickness **610**. Center thickness **610** can be between 3 and 8 mm thick. In some embodiments, center thickness **610** is between 5 and 6 mm thick, such as 5.47 mm. Edge thickness **611** represents the thickness of domed lens **402** around the entirety of its circumference. Edge thickness **611** can be between 1 and 2 mm, such as 1.3 mm. Film thickness **612** can represent the total thickness of OCA layer **404**, tinted ink layer **406**, and mirror film layer **408**. Film thickness **612** can be between 0.1 mm and 0.2 mm. In some embodiments, film thickness **612** is 0.17 mm. Lens assembly **212** can have a diameter **613** of between 75-120 mm. In some embodiments, lens assembly **212** has diameter **613** of between 90-100 mm. As a specific example, diameter **613** can be 93.2 mm.

[0064] FIG. **6B** illustrates a cross section **600B** of an embodiment of lens assembly **212**. Cross section **600B** of FIG. **6B** is not to scale. Cross section **600B** is indicated in FIG. **2B**. Cross section **600B** only represents lens assembly **212**. Ambient light sensor **610** has field-of-view **620**. The domed shape of domed lens **402** causes field-of-view **620** to be expanded to field-of-view **630** at surface **501**. This increase in the angle of field-of-view **630** compared with field-of-view **620** is caused at least in part by the domed shape of domed lens **402**. This same effect can make the electronic display of the smart thermostat **200** visible at a greater angle to a user. This effect can be particularly useful where a user may be approaching smart thermostat **200** from the side, such as when smart thermostat **200** is mounted to a wall of a hallway.

[0065] In order to create the embodiments of lens assembly **122**, **212**, and **400** detailed herein, a method of manufacturing can be performed. FIG. **7** illustrates an embodiment of a method **700** for manufacturing lens assemblies as detailed herein, such as for installation in a smart thermostat or other smart device. Specifically, method **700** allows for the polarization of the mirror film layer of the lens assembly to be accounted for during manufacture to ensure that light emitted from an electronic display of the smart thermostat or smart device is not blocked. Light emitted from such an electronic display can have a particular polarization and if the mirror film's polarization is normal to the polarization of the electronic display, all or a significant portion of emitted light can be blocked by the mirror film.

[0066] At block **710**, injection compression molding can be used to form the domed lens. In injection compression molding, a liquified material, such as PMMA, can be injected into a mold that has not been completely closed. Following the material being injected or while the material is being injected, the mold may be completely closed. Excess material may be squeezed out of the mold where the two halves of the mold meet. This additional material may be removed, such as via a machining process. The resulting domed lens may be defect-free or nearly defect-free having highly smooth surfaces.

[0067] Mirror film may be acquired in the form of a roll of film material. Throughout the roll, polarization of the mirror film may vary. For example, at a beginning of a mirror film roll, polarization may be close to vertical, but may transition to near horizontal further into the roll. Therefore, if the polarization direction is not determined, significant variability can result from installing a mirror film in relation to an electronic display. At block **720**, cutouts, such as circular cutouts, of mirror film can be made from a mirror film sheet, which can be in the form of a mirror film roll.

[0068] At block **730**, for each mirror film cutout, the direction of polarization can be determined. In some embodiments, this can be accomplished by shining polarized light at the mirror film cutout

and determining an orientation that either maximizes obstruction or minimizes obstruction of the light.

[0069] At block **740**, based on the measured polarization of block **730**, an orientation at which the mirror film cutout is to be installed in relation to the electronic display is determined. The direction of polarization of the light emitted by the electronic display is known; therefore, the mirror film's orientation is adjusted to account for the known polarization of the electronic display. At some stage in the manufacturing process, the mirror film cutout is adhered to the domed lens along with any intervening layers. Since these layers may be symmetrical (e.g., circular with no holes or discontinuities), the orientation of the mirror film cutout may not matter. In some embodiments, the colored ink layer is applied to the mirror film. The OCA layer is then attached to the ink/mirror film, after which the OCA/ink/mirror film may be attached with the domed lens. The masking layer may be asymmetric, such as due to a cutout for an ambient light sensor; therefore, orientation of the mirror film with respect to the masking layer needs to take into account the polarization. At block **750**, the masking layer is affixed with the mirrored film cutout using the determined orientation. Block **750** can be performed before or after the mirror film layer has been affixed to the domed lens (and any intervening layers). Following block **750**, the domed lens, mirror film cutout, and masking layer and possibly other layers now form a lens assembly, such as lens assembly **400**. Additionally, one or more adhesive layers (e.g., adhesive layer **412** and adhesive layer **414**) may be added to the lens assembly for use in attaching the lens assembly to the smart thermostat or smart device.

[0070] At block **760**, the lens assembly is installed in the smart thermostat or smart device housing according to the determined orientation to minimize the polarizing effect of the mirror film on light emitted by the electronic display. For example, if the light emitted by the electronic display is vertically polarized, the orientation of the mirror film in the lens assembly allows for transmission of vertically polarized light. In the context of block **760**, “minimize” can mean approximately keep to a minimum, such as block less than 5%, 10%, or 20% of the emitted light of the electronic display due to the polarizing effect of the mirror film cutout.

[0071] It should be noted that the methods, systems, and devices discussed above are intended merely to be examples. It must be stressed that various embodiments may omit, substitute, or add various procedures or components as appropriate. For instance, it should be appreciated that, in alternative embodiments, the methods may be performed in an order different from that described, and that various steps may be added, omitted, or combined. Also, features described with respect to certain embodiments may be combined in various other embodiments. Different aspects and elements of the embodiments may be combined in a similar manner. Also, it should be emphasized that technology evolves and, thus, many of the elements are examples and should not be interpreted to limit the scope of the invention.

[0072] Specific details are given in the description to provide a thorough understanding of the embodiments. However, it will be understood by one of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, well-known, processes, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the embodiments. This description provides example embodiments only, and is not intended to limit the scope, applicability, or configuration of the invention. Rather, the preceding description of the embodiments will provide those skilled in the art with an enabling description for implementing embodiments of the invention. Various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the invention.

[0073] Also, it is noted that the embodiments may be described as a process which is depicted as a flow diagram or block diagram. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure.

[0074] Having described several embodiments, it will be recognized by those of skill in the art that

various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the invention. For example, the above elements may merely be a component of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered. Accordingly, the above description should not be taken as limiting the scope of the invention.

Claims

1. A thermostat, comprising: a housing; an electronic display housed by the housing; an ambient light sensor housed by the housing; a lens assembly attached with the housing, the lens assembly comprising: a circular lens, wherein: a first surface of the circular lens is flat; a perimeter of the circular lens is defined by a second surface that is perpendicular to the first surface of the circular lens; a third surface of the circular lens is domed; and the ambient light sensor measures ambient light within an ambient environment of the thermostat through the circular lens.
2. The thermostat of claim 1, wherein a size of a field of view of the ambient light sensor is increased by the third surface of the circular lens.
3. The thermostat of claim 1, wherein the lens assembly further comprises a polarized mirrored film attached with the circular lens.
4. The thermostat of claim 3, wherein the polarized mirrored film is oriented with respect to the electronic display to increase transmissivity of polarized light emitted by the electronic display.
5. The thermostat of claim 4, wherein the lens assembly further comprises a colored ink layer attached between the polarized mirrored film and the circular lens.
6. The thermostat of claim 5, wherein the lens assembly further comprises a masking layer attached between the polarized mirrored film and the electronic display.
7. The thermostat of claim 5, wherein the lens assembly further comprises one or more adhesive layers and only the one or more adhesive layers secure the lens assembly to the thermostat.
8. The thermostat of claim 2, further comprising a radar sensor that emits radar signals and detects reflections of the radar signals through the lens assembly.
9. The thermostat of claim 8, wherein the circular lens comprises polymethyl methacrylate (PMMA).
10. The thermostat of claim 9, wherein the circular lens is formed using injection compression molding.
11. The thermostat of claim 10, wherein the second surface of the circular lens is completely surrounded by a rotatable ring.
12. A method for manufacturing a thermostat lens assembly, the method comprising: creating a domed lens, wherein: a first surface of the circular lens is flat; a perimeter of the circular lens is defined by a second surface that is perpendicular to the first surface of the circular lens; and a third surface of the circular lens is domed; measuring a direction of polarization of a mirror film; determining an orientation of the mirror film to attach with a lens assembly comprising the domed lens based on the measured direction of polarization of the mirror film; and installing the lens assembly in a thermostat such that the determined orientation minimizes a polarizing effect on light emitted by an electronic display of the thermostat.
13. The method for manufacturing the thermostat lens assembly of claim 12, further comprising affixing a masking layer with the mirror film based on the determined orientation, wherein the masking layer is asymmetrical.
14. The method of claim 12, wherein the domed lens is created using injection compression molding.
15. The method of claim 14, wherein the domed lens is created from polymethyl methacrylate (PMMA).

16. The method for manufacturing the thermostat lens assembly of claim 13, further comprising: affixing the circular lens to a colored ink layer using an optically clear adhesive; and affixing the mirrored film with the colored ink layer.

17. The method for manufacturing the thermostat lens assembly of claim 16, further comprising: affixing one or more adhesive layers to the masking layer, wherein only the one or more adhesive layers secure the lens assembly to the thermostat.

18. A thermostat lens assembly, comprising: a circular lens, wherein: a first surface of the circular lens is flat; a perimeter of the circular lens is defined by a second surface that is perpendicular to the first surface of the circular lens; and a third surface of the circular lens is domed.

19. The thermostat lens assembly of claim 18, further comprising: a mirrored film layer having a polarization; and a masking layer that is asymmetrical, wherein the mirrored film is oriented with respect to the masking layer based on the polarization.

20. The thermostat lens assembly of claim 19, further comprising: an optically clear adhesive layer; and a colored ink layer, wherein the optically clear adhesive layer adheres the lens to the colored ink layer and the mirrored film layer is in direct contact with the colored ink layer.
